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DATA STURCTURE AND ALGORITHM (LAB EXERCISES AND PRACTICE)

OUTPUT TRACING QUESTION

PSEUDO CODE (TRACING)

Based on the following pseudo code in **Figure 1**, complete the trace table given in **Table 1**.

[5 marks]

```
1.  START
2.  READ n, m
3.  IF (n > = m)
    3.1 START_IF
        3.1.1   IF (n > 10)
            3.1.1.1 START_IF
                3.1.1.1.1   IF (m> 10)
                    3.1.1.1.1.1 START_IF
                        3.1.1.1.1.1.1     PRINT "both n and m is greater than 10"
                    3.1.1.1.1.2 END_IF
                3.1.1.1.2   IF (n = = m)
                    3.1.1.1.2.1 START_IF
                        3.1.1.1.2.1.1.1 PRINT "n is equal to m"
                    3.1.1.1.2.2 END_IF
            3.1.1.2 END_IF
    3.2 END_IF
4.  ELSE
    4.1 PRINT (n-m)*2
5.  PRINT n, m
6.  END
```

Figure 1

ANSWER

Table 1

n	m	Output
0	**0**	0 0
10	**0**	10 0
20	**10**	20 10
20	**20**	both n and m is greater than 10 n is equal to m 20 20
0	**10**	-20 0 10

PSEUDO CODE (TRACING)

Based on the following pseudo code in **Figure 2**, complete the trace table given in **Table 2**.

[9 marks]

Data to be used for this problem is:

noItems

3

itemID	**CP**	**SP**	**units**
1111	32	40	10
2222	25	30	5
3333	57	65	2

```
1. START
2. SET Total to 0
3. READ noItems
4. WHILE (noItems IS NOT 0)
   4.1 START_WHILE
       4.1.1  READ itemID, CP, SP, units
       4.1.2  P = (SP * units) - (CP * units)
       4.1.3  ADD P to Total
       4.1.4  PRINT itemID, P
       4.1.5  SUBTRACT 1 from noItems
   4.2 END_WHILE
5. PRINT Total
6. END
```

Figure 2

ANSWER:

Table 2

Total	noItems	P	Output

0 80 105 121	3 2 1 0	80 25 16	1111,80 2222,105 1111,121 121

FLOWCHART (TRACING)

Use the flowchart in **Figure 3** to answer Question 3. **[10 marks]**

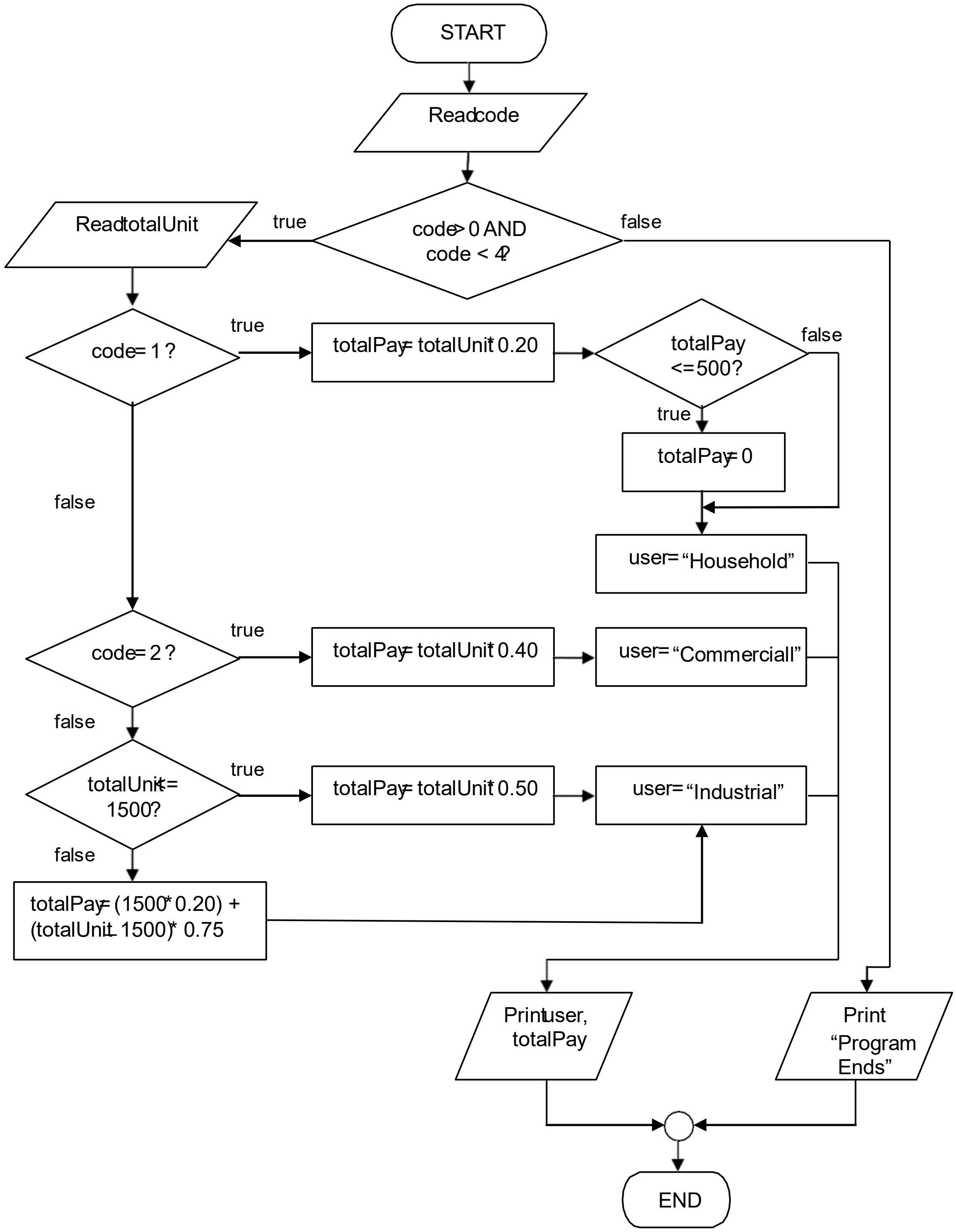


Figure 3

Trace the variables of the flowchart in **Figure 3** using the following input values:

	Code	totalUnit
i.	0	200
ii.	1	375

iii. 2 550

iv. 3 4500

v. 1 30

vi. 2 1000

vii. 3 1200

ANSWER:

Fill your answers in the **Table 3** provided below.

Table 3

code	totalPay	Output
0		Program Ends
1	75	Household, 0
2	220	Commercial,220
3	2250	Industrial,2250
1	6	Household,0
2	400	Commercial,400
3	600	Industrial,600

FLOWCHART (TRACING)

Trace the flowchart below in **Figure 4** by fill in the table given in **Table 4.** **[15 marks]**

!

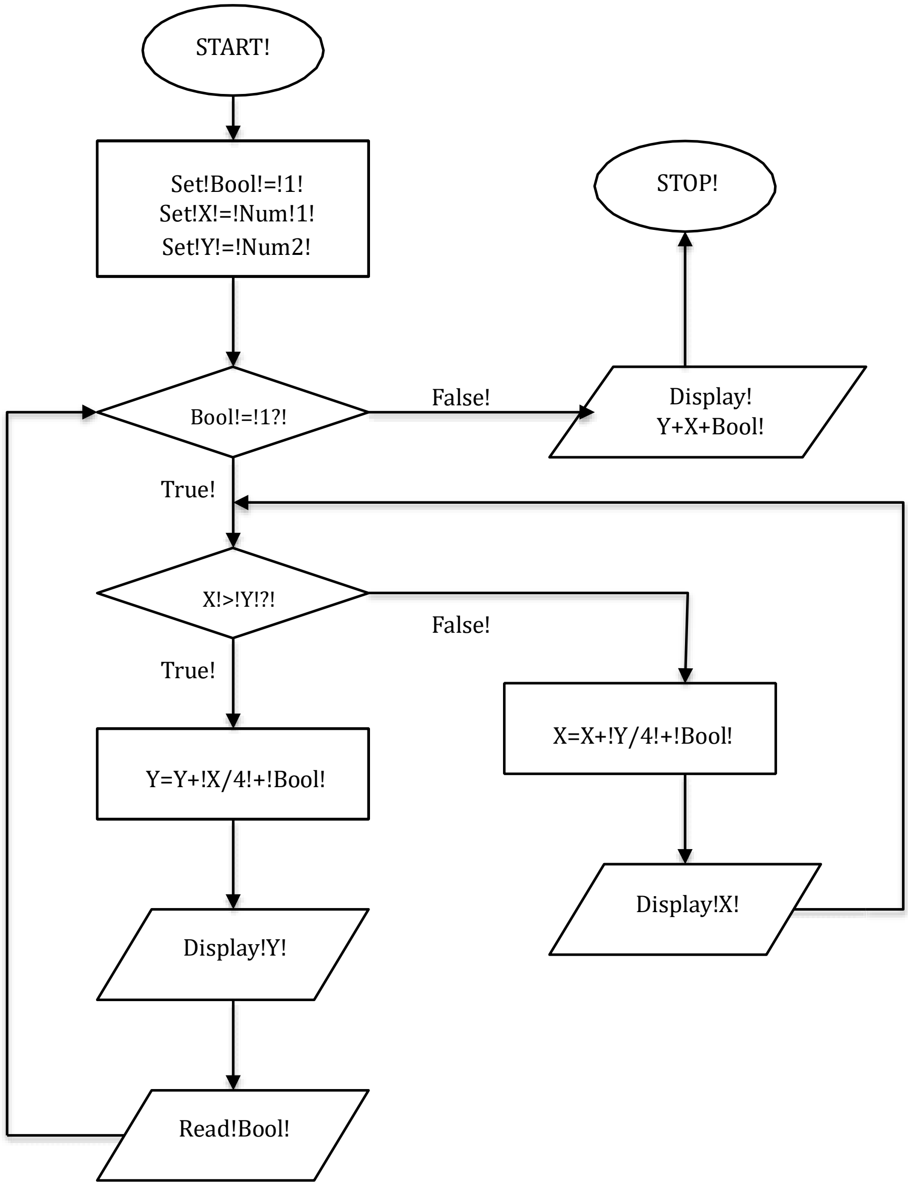


Figure 4

ANSWER:

Table 4

Num1	Num2	Bool	X	Y	Output
7	8	1 1 2			

23	5	1 1 -5			
32	23	1 1 1 10			

PROBLEM SOLVING QUESTION

PROVIDE COMPLETE PSEUDO CODE

[20 marks]

Write a pseudo code for a program that will implement the following decision table in **Table 5**. The program will print the input grade point and the class of degree based on a user input. The program will terminate the loop when a user input a sentinel value other than ‘y’ or ‘Y’. **Table 5**

GRADE POINT	Class of Degree
0.0 – 0.99	Failed
1.0 – 2.00	General degree
2.1 – 2.7	Second class lower
2.71 – 3.69	Second class upper
3.7 – 4.00	First Class

ANSWER:

```
1. START

2. DECLARE grade_point AS FLOAT
3. DECLARE user_input AS CHAR

4. DO
    4.1  PRINT "Enter your grade point (0.0 to 4.0): "
    4.2  READ grade_point
    4.3  PRINT grade_point

    4.4  IF grade_point >= 0.0 AND grade_point <= 0.99 THEN
             4.4.1 PRINT "Failed"

    4.5  ELSE IF grade_point >= 1.0 AND grade_point <= 2.00 THEN
             4.5.1 PRINT "General degree"

    4.6  ELSE IF grade_point >= 2.1 AND grade_point <= 2.7 THEN
             4.6.1 PRINT "Second class lower"

    4.7  ELSE IF grade_point >= 2.71 AND grade_point <= 3.69 THEN
             4.7.1 PRINT "Second class upper"

    4.8  ELSE IF grade_point >= 3.7 AND grade_point <= 4.00 THEN
             4.8.1 PRINT "First Class"

    4.9  ELSE
             4.9.1 PRINT "Invalid grade point"
    4.10 END IF
```

4.11 PRINT "Do you want to input another grade point? (y/Y to continue, any other key to exit): "
4.12 READ user_input

5. WHILE user_input == 'y' OR user_input == 'Y'

6. PRINT "Program terminated."
7. END

PROVIDE COMPLETE PROGRAM

[30 marks]

Mode Auto Dealer is a used car business owned by Mr. Muhammad. He employs you to develop a program that will calculate the sales discount to be applied to a vehicle, based on its year of manufacture and type. The discount is extracted from a two-dimensional table as in **Table 6** below:

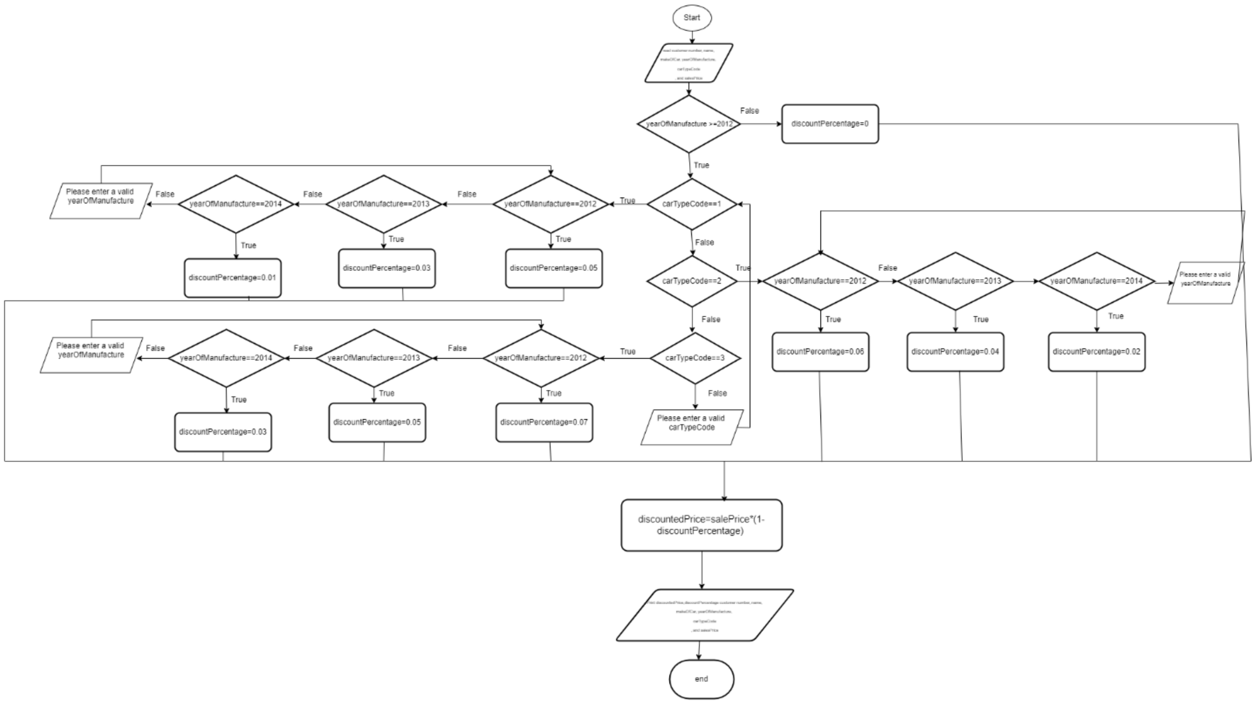
Table 6

Year of manufacture	Discount Percentage		
	Small	Medium	Luxury
	1	2	3
2012	0.05	0.06	0.07
2013	0.03	0.04	0.05
2014	0.01	0.02	0.03

The year of manufacture of the vehicle is divided into three categories (2012, 2013 and 2014), and the type of car is divided into three categories (small, medium and luxury). No discount is given for a vehicle older than 2012. Your program is to read the vehicle file, which contains the customer number and name, the make of car, year of manufacture, car type code (1, 2, or 3) and the sales price. Use the year of manufacture and the car type code as guidelines to retrieve the discount percentage table, then apply the discount percentage to the sales price to determine the discounted price of the vehicle. Print all the details including discounted price. Before writing the program you are required to plan your problem solving steps in a flowchart(s).

Note: Additional papers will be given to answer this question.

ANSWER


Start
yearOfManufacture >=2012
False
discountPercentage=0
True
carTypeCode==1
yearOfManufacture==2012
yearOfManufacture==2013
yearOfManufacture==2014
Please enter a valid yearOfManufacture
discountPercentage=0.05
discountPercentage=0.03
discountPercentage=0.01
carTypeCode==2
yearOfManufacture==2012
yearOfManufacture==2013
yearOfManufacture==2014
Please enter a valid yearOfManufacture
discountPercentage=0.06
discountPercentage=0.04
discountPercentage=0.02
carTypeCode==3
yearOfManufacture==2012
yearOfManufacture==2013
yearOfManufacture==2014
Please enter a valid yearOfManufacture
discountPercentage=0.07
discountPercentage=0.05
discountPercentage=0.03
Please enter a valid carTypeCode
discountedPrice=salePrice*(1-discountPercentage)
end

```cpp
#include <iostream>
#include <vector>
#include <string>
#include <iomanip>

using namespace std;

struct Vehicle {
    string customerNumber;
    string name;
    string make;
    int year;
    int carTypeCode;
    double salesPrice;
    double discountedPrice;
};

double calculateDiscount(int carTypeCode, int year, double salesPrice) {
    double discountPercentage = 0.0;


    if (year < 2012) {
        discountPercentage = 0.0;
    } else {

        if (carTypeCode == 1) {
            if (year == 2012) discountPercentage = 0.05;
            else if (year == 2013) discountPercentage = 0.03;
            else if (year == 2014) discountPercentage = 0.01;
        } else if (carTypeCode == 2) {
            if (year == 2012) discountPercentage = 0.06;
            else if (year == 2013) discountPercentage = 0.04;
            else if (year == 2014) discountPercentage = 0.02;
        } else if (carTypeCode == 3) {
            if (year == 2012) discountPercentage = 0.07;
            else if (year == 2013) discountPercentage = 0.05;
            else if (year == 2014) discountPercentage = 0.03;
        }
    }

    return salesPrice * (1 - discountPercentage);
}

Vehicle inputVehicleDetails() {
    Vehicle v;

    cout << "Enter customer number: ";
    cin.ignore();
    getline(cin, v.customerNumber);

    cout << "Enter customer name: ";
    getline(cin, v.name);

    cout << "Enter make of car: ";
    getline(cin, v.make);

    while (true) {
        cout << "Enter year of manufacture (2012, 2013, 2014): ";
        cin >> v.year;
        if (v.year == 2012 || v.year == 2013 || v.year == 2014 || v.year<2012) {
            break;
        } else {
            cout << "Invalid year. Please enter 2012, 2013, or 2014.\n";
        }
    }

    while (true) {
        cout << "Select car type:\n";
        cout << "1. Small\n";
        cout << "2. Medium\n";
        cout << "3. Luxury\n";
        cout << "Enter car type code (1-3): ";
        cin >> v.carTypeCode;
        if (v.carTypeCode == 1 || v.carTypeCode == 2 || v.carTypeCode == 3) {
            break;
        } else {
            cout << "Invalid car type code. Please enter 1, 2, or 3.\n";
        }
    }

    cout << "Enter sales price: ";
    cin >> v.salesPrice;

    v.discountedPrice = calculateDiscount(v.carTypeCode, v.year, v.salesPrice);

    return v;
}
```

```cpp
int main() {
    int numVehicles;
    cout << "Enter the number of vehicles to input: ";
    cin >> numVehicles;

    vector<Vehicle> vehicles;

    for (int i = 0; i < numVehicles; ++i) {
        cout << "\nInput details for vehicle " << (i + 1) << ":\n";
        vehicles.push_back(inputVehicleDetails());
    }

    cout << fixed << setprecision(2);
    cout << "\nCustomer Number | Name | Make | Year | Car Type Code | Sales Price | Discounted Price\n";
    cout << "------------------------------------------------------------------------------------------\n";

    for (const auto& vehicle : vehicles) {
        cout << vehicle.customerNumber << " | "
             << vehicle.name << " | "
             << vehicle.make << " | "
             << vehicle.year << " | "
             << vehicle.carTypeCode << " | "
             << "$" << vehicle.salesPrice << " | "
             << "$" << vehicle.discountedPrice << "\n";
    }

    return 0;
}
```

```
Enter the number of vehicles to input: 1

Input details for vehicle 1:
Enter customer number: 01234567890
Enter customer name: LING YU QIAN
Enter make of car: tOYOTA
Enter year of manufacture (2012, 2013, 2014): 2011
Select car type:
1. Small
2. Medium
3. Luxury
Enter car type code (1-3): 1
Enter sales price: 100000

Customer Number | Name | Make | Year | Car Type Code | Sales Price | Discounted Price
------------------------------------------------------------------------------------------
01234567890 | LING YU QIAN | tOYOTA | 2011 | 1 | $100000.00 | $100000.00

--------------------------------
Process exited after 46.91 seconds with return value 0
Press any key to continue . . .
```